
Lecture Notes

on Eigenvectors and Stability

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1 Normalized Eigenvector

If

$$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix},$$

then the **normalized eigenvector** is written as

$$\hat{v} = \begin{pmatrix} \frac{v_1}{v_1+v_2} \\ \frac{v_2}{v_1+v_2} \end{pmatrix}.$$

👉 **Example:** The normalized eigenvector of

$$\vec{v} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

is given by

$$\hat{v} = \begin{pmatrix} \frac{2}{5} \\ \frac{3}{5} \end{pmatrix}.$$

2 Equilibrium or Stability

Suppose in a model, each day 10% of the susceptibles become infected and 20% of the infected recover and become susceptible again. **Question:** What is the transfer matrix?

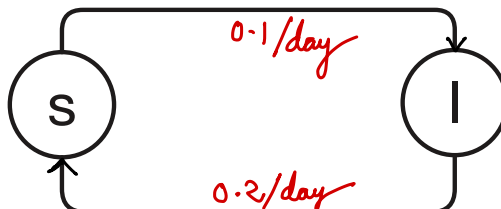
Further, let, at $t = 0$,

$$x(0) = \begin{pmatrix} S(0) \\ I(0) \end{pmatrix} = \begin{pmatrix} 297 \\ 3 \end{pmatrix}.$$

Then, what happens after a long time (say, 50 days or more)? How many individuals are susceptible and how many are infected at equilibrium? What happens if you change the initial condition to

$$x(0) = \begin{pmatrix} 10 \\ 290 \end{pmatrix}?$$

Note: The total population initially is still $10 + 290 = 300$.



let S and I denote susceptible and infected individuals at time t .

$$S(t), I(t)$$

let A be the transfer matrix,

$$A = \begin{bmatrix} S(t) & I(t) \\ 1-0.1 & 0.2 \\ 0.1 & 1-0.2 \end{bmatrix} \begin{matrix} S(t+1) \\ I(t+1) \end{matrix} = \begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix}$$

Given, $x(0) = \begin{pmatrix} 297 \\ 3 \end{pmatrix} \quad x(t)$

$$x(1) = Ax(0) = \begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix} \begin{bmatrix} 297 \\ 3 \end{bmatrix} = \dots$$

$$x(2) = Ax(1) = A(Ax(0)) = A^2 x(0)$$

$$x(3) = Ax(2) = A(A^2 x(0)) = A^3 x(0)$$

$\vdots$

$$x(n) = A^n x(0)$$

Q: for $n=50$ or, $n > 50$, what is $x(n)$?

$$x(50) = A^{50} x(0) = \begin{bmatrix} 200 \\ 100 \end{bmatrix}.$$

$$x(51) = A(x(50)) = \begin{bmatrix} 200 \\ 100 \end{bmatrix}.$$

Eigenvalues of the matrix A are $\lambda_1 = 1$, $\lambda_2 = 7/10$.

So the dominant eigenvalue is 1.

Then find the eigenvector corresponding to the eigenvalue, i.e., $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$.

The corresponding normalized eigenvector is $\vec{v} = \begin{pmatrix} 2/3 \\ 1/3 \end{pmatrix}$.

$$\text{So, } 300 \begin{pmatrix} 2/3 \\ 1/3 \end{pmatrix} = \begin{pmatrix} 200 \\ 100 \end{pmatrix}.$$

→ total initial population.

Thus there are 200 susceptible and 100 infected individuals at equilibrium.

Note that even we change the initial condition $x(0) = \begin{pmatrix} 10 \\ 290 \end{pmatrix}$, the population distribution is going to be

$$300 \begin{pmatrix} 2/3 \\ 1/3 \end{pmatrix} = \begin{pmatrix} 200 \\ 100 \end{pmatrix},$$

i.e, same. In a summary, there will be no change of the population distribution at equilibrium until total initial population 300 and the transfer matrix are unchanged.

2.1 Steps to Find the Equilibrium (or Steady State)

1. Find the **transfer matrix**.
2. Compute all the **eigenvalues** and identify the **dominant eigenvalue**.
3. Determine the **eigenvector** corresponding to the dominant eigenvalue.
4. Evaluate the corresponding **normalized eigenvector**.
5. To obtain the final result, multiply this normalized eigenvector by the total initial population.

Note: If the initial condition is changed to

$$x(0) = \begin{pmatrix} 10 \\ 290 \end{pmatrix}$$

instead of

$$x(0) = \begin{pmatrix} 297 \\ 3 \end{pmatrix},$$

while keeping the total population fixed at 300, the populations of each compartment at equilibrium will remain unchanged.

Example:

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}.$$

Find all eigenvalues and the corresponding eigenvectors.

Eigenvalues

$$\Rightarrow \det(A - \lambda I) = 0$$

$$\Rightarrow \begin{vmatrix} 2-\lambda & 1 \\ 0 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (2-\lambda)(2-\lambda) - 1 \times 0 = 0.$$

$$\Rightarrow (2-\lambda)^2 = 0 \Rightarrow \lambda = 2, 2.$$

Let $\vec{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ be the eigenvector corresponding to $\lambda = 2$.

$$\begin{bmatrix} 2-2 & 1 \\ 0 & 2-2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

$$\Rightarrow \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} (0 \times v_1) + (1 \times v_2) \\ (0 \times v_1) + (0 \times v_2) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

$$\Rightarrow \begin{pmatrix} v_2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

$$\Rightarrow v_2 = 0.$$

So v_1 is arbitrary, and let $v_1 = 1$.

$$\therefore \vec{v} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

$$(A - 2I)\vec{v} = \vec{0}$$

$$\Rightarrow \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} - 2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} - \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2-2 & 1-0 \\ 0-0 & 2-2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Example:

$$A = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix}.$$

Find all eigenvalues and the corresponding eigenvectors.

$$\det \begin{bmatrix} 2-\lambda & 1 \\ 2 & 1-\lambda \end{bmatrix} = 0 \Rightarrow (2-\lambda)(1-\lambda) - (1 \times 2) = 0.$$

$$\textcircled{\text{OR}} \quad \begin{vmatrix} 2-\lambda & 1 \\ 2 & 1-\lambda \end{vmatrix} = 0 \Rightarrow 2 - 2\lambda - \lambda + \lambda^2 - 2 = 0.$$

$$\Rightarrow \lambda^2 - 3\lambda = 0.$$

$$\Rightarrow \lambda(\lambda - 3) = 0.$$

$$\therefore \lambda_1 = 0, \lambda_2 = 3.$$

For $\lambda_1 = 0$, let $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ be the eigenvector.

$$\begin{bmatrix} 2-0 & 1 \\ 2 & 1-0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

$$\Rightarrow \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2x_1 + x_2 \\ 2x_1 + x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow 2x_1 + x_2 = 0. \Rightarrow x_2 = -2x_1$$

$$\text{let } x_1 = 1, \therefore x_2 = -2. \therefore \vec{x} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

For $\lambda = 3$, let $\vec{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$

$$\begin{bmatrix} 2-3 & 1 \\ 2 & 1-3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -y_1 + y_2 \\ 2y_1 - 2y_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow -y_1 + y_2 = 0 \Rightarrow y_1 = y_2 \text{ --- (1)}$$

$$2y_1 - 2y_2 = 0 \Rightarrow 2y_1 = 2y_2 \Rightarrow y_1 = y_2 \text{ --- (2)}$$

(Note: we have same equations (1) & (2))

$$\text{Let } y_1 = 1, \therefore y_2 = 1.$$

$$\therefore \vec{y} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$